

BALANCED SPANNING TREES FOR TRIANGULAR STRIP LATTICES

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ABSTRACT. A balanced spanning tree is a spanning tree that contains an edge whose removal partitions the vertices into exactly two connected subtrees of equal size. In this paper, we establish explicit recurrence relations for the number of spanning trees in $2 \times n$ *triangular strip lattices*—obtained by adding a diagonal edge to each square of a $2 \times n$ grid graph—generalizing combinatorial counting techniques introduced by Raff [Raf08]. We then adapt arguments of Gallagher and Tapp [GT25] to count balanced spanning trees of arbitrary triangular strip lattices. We establish sharp asymptotic bounds for the proportion of balanced spanning trees as $n \rightarrow \infty$. Finally, we determine the probability that a spanning tree of a $2 \times n$ triangular strip lattice chosen uniformly at random is balanced as $n \rightarrow \infty$.

1. INTRODUCTION

Balanced spanning trees (Definition 1) have recently attracted attention because of their applications to computational redistricting. One of the most prominent redistricting algorithms is ReCom, which has been observed empirically to generate districting plans with compact districts [DDS21, CBH⁺21]. More recently, Procaccia and Tucker-Foltz [PTF22] provided a theoretical explanation for this phenomenon by showing that the spanning-tree-based sampling procedure used by ReCom favors redistricting plans with shorter boundaries, which are more compact. This connection between spanning trees and compact districts is rooted in the way ReCom constructs new districting plans.

Specifically, ReCom operates on the dual graph of a geographic region, where each vertex represents a geographic unit (such as a census block, precinct, or county) and edges connect adjacent geographic units (see Figure 1). At each iteration, the algorithm temporarily merges two adjacent districts, samples a spanning tree of the corresponding subgraph, and removes an edge whose deletion yields two connected components with nearly equal populations; that is, it seeks balanced spanning trees in the sense of Definition 1 [DDS21]. The two components then become the new districts. Consequently, understanding the prevalence of balanced spanning trees may provide theoretical insight into the behavior of ReCom and related spanning-tree-based redistricting algorithms.

As illustrated in Figure 1, dual graphs arising from real geographic regions often contain square and triangular faces. This observation has motivated several recent studies of spanning trees and balanced partitions in families of graphs where the faces are all squares or triangles. Cannon, Pegden, and Tucker-Foltz [CPTF24] studied balanced partitions of grid graphs and developed polynomial-time algorithms for sampling balanced forests, establishing bounds on the probability that a random spanning tree admits a balanced partition. Gallagher and Tapp [GT25] later derived an exact expression for the proportion of balanced spanning trees in the $2 \times n$ grid graph. Wang [Wan23] investigated spanning tree enumeration on three families of graphs with triangular faces—rows of triangles, fan graphs, and chains of triangles. More recently, Gugliani [Gug25]



FIGURE 1. Dual graph for West Virginia where the geographical units are counties.

established a lower bound for the proportion of balanced spanning trees in hexagonal regions of triangular lattices.

In this paper, we study balanced spanning trees in rows of triangles. In particular, we consider $2 \times n$ *triangular strip lattices*, obtained by adding a diagonal to each square face of a $2 \times n$ grid graph, and establish sharp asymptotic bounds for the proportion of balanced spanning trees as $n \rightarrow \infty$. We further determine the limiting probability that a uniformly random spanning tree of a $2 \times n$ triangular strip lattice is balanced.

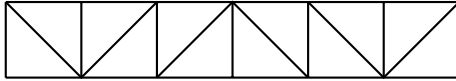


FIGURE 2. An example of a 2-by-7 triangular strip lattice in $\mathcal{G}_{14}$.

Before presenting our main theorems, we formally define a balanced spanning tree.

Definition 1 (Balanced spanning tree). Let $\mathcal{T}$ be a spanning tree of a graph $G = (V, E)$ with an even number of vertices, $|V| = 2n$. We say that $\mathcal{T}$ is a *balanced spanning tree* if there exists an edge $e \in E(\mathcal{T})$ such that the forest $\mathcal{T} \setminus \{e\}$ consists of exactly two disjoint, connected subtrees, each spanning exactly n vertices in G .

Our main results are summarized in the following two theorems.

Theorem 1.1 (Asymptotic Bounds for Balanced Spanning Trees). *Let $\mathcal{G}_{2n}$ be the family of triangular strip lattices with $2n$ vertices, where $n \in \mathbb{N}$. Then for any $\Gamma_{2n} \in \mathcal{G}_{2n}$, the asymptotic proportion*

of balanced spanning trees satisfies the following bounds:

$$\text{Even Case } (n = 2m): \quad 0.63087 \leq \lim_{n \rightarrow \infty} \frac{S_{2n}}{T_{2n}} \leq 0.668328,$$

$$\text{Odd Case } (n = 2m + 1): \quad 0.544933 \leq \lim_{n \rightarrow \infty} \frac{S_{2n}}{T_{2n}} \leq 0.668328.$$

where T_{2n} and S_{2n} denote the number of spanning trees and balanced spanning trees respectively in Γ_{2n} .

While Theorem 1.1 provides sharp asymptotic upper and lower bounds, we are also interested in the behavior of a typical triangular strip lattice. To model such a lattice, we assign each diagonal edge one of the two possible orientations independently and with equal probability (see Section 4 for a more precise description), where the orientation specifies whether the diagonal joins the top-left and bottom-right vertices or the top-right and bottom-left vertices of its corresponding square face. We then calculate the asymptotic proportion of balanced spanning trees for this random triangular strip lattice.

Theorem 1.2 (Average Proportion for Typical Triangular Strip Lattices). *Let Γ_{2n} be a triangular strip lattice whose diagonal configuration is chosen uniformly at random. As $n \rightarrow \infty$, the expected proportion of balanced spanning trees converges to the mean of the asymptotic bounds:*

$$\text{Even Case } (n = 2m): \quad \lim_{n \rightarrow \infty} \mathbb{E} \left[\frac{S_{2n}}{T_{2n}} \right] \approx 0.6496,$$

$$\text{Odd Case } (n = 2m + 1): \quad \lim_{n \rightarrow \infty} \mathbb{E} \left[\frac{S_{2n}}{T_{2n}} \right] \approx 0.6066.$$

ACKNOWLEDGMENTS

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2. COUNTING SPANNING TREES

In this section, we count the number of spanning trees in a given triangular strip lattice. Our approach is inspired by Raff's method for enumerating spanning trees in grid graphs [Raf08].

2.1. Counting spanning trees of a triangular strip lattice with an even number of vertices. We begin by considering triangular strip lattices with an even number of vertices, as balanced spanning trees are defined only for graphs with an even number of vertices. Let Γ_{2n} denote an arbitrary triangular strip lattice with $2n$ vertices, where $n \in \mathbb{N}$. Let T_{2n} denote the number of spanning trees of Γ_{2n} .

Theorem 2.1 (Number of spanning trees in Γ_{2n}). *For any triangular strip lattice Γ_{2n} , the number of spanning trees satisfies the recurrence*

$$T_{2(n+2)} = 7T_{2(n+1)} - T_{2n},$$

with initial conditions $T_2 = 1$ and $T_4 = 8$.

Proof. To derive a recurrence for the number of spanning trees, we will examine all ways a spanning tree in $\Gamma_{2(n+1)}$ can be constructed by adding two vertices to the right end of Γ_{2n} .

Let u and v denote the two rightmost vertices of $\Gamma_{2(n+1)}$, and let $\mathcal{T}$ be a spanning tree of $\Gamma_{2(n+1)}$. Since $\mathcal{T}$ is a spanning tree of $\Gamma_{2(n+1)}$, the graph $\mathcal{T} \setminus \{u, v\}$ contains no loops and spans all vertices of Γ_{2n} . Consequently, $\mathcal{T} \setminus \{u, v\}$ is either

- (1) a spanning tree of Γ_{2n} or
- (2) a spanning forest of Γ_{2n} consisting of two components, with the two rightmost vertices of Γ_{2n} lying in different trees.

Note that it suffices to consider forests from (2) because if there are more than two trees in the forest or if the rightmost vertices of the forest lie on the same tree, the added vertices cannot adjoin the trees into a single spanning tree.

Accordingly, let $\mathcal{T}_{2n}$ denote the set of spanning trees in Γ_{2n} , and let $\mathcal{F}_{2n}$ denote the set of spanning forests in Γ_{2n} described in (2). Starting from a spanning tree in $\mathcal{T}_{2n}$, we can extend to a spanning tree in $\mathcal{T}_{2(n+1)}$ by adding two new vertices u and v to its right end. For this to happen, we must pick edges that connect both u and v to the rest of the tree without creating any loops. Note that if one were to pick a diagonal, that diagonal must either join the top-left and bottom-right vertices or the top-right and bottom-left vertices as predetermined by the structure of Γ_{2n} . For either type of diagonal, there are five possible ways to extend a spanning tree in $\mathcal{T}_{2n}$ to a spanning tree in $\mathcal{T}_{2(n+1)}$.

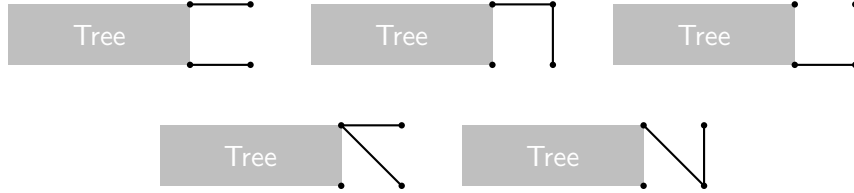


FIGURE 3. Possible ways to extend a spanning tree in $\mathcal{T}_{2n}$ to a spanning tree in $\mathcal{T}_{2(n+1)}$.

The only structure that can extend to a tree in $\mathcal{T}_{2(n+1)}$ in addition to spanning trees is a forest in $\mathcal{F}_{2n}$. There are three ways each forest in $\mathcal{F}_{2n}$ can be extended to a tree in $\mathcal{T}_{2(n+1)}$ (See Figure 4).

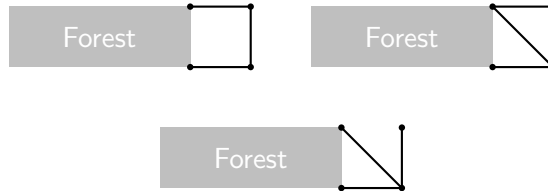


FIGURE 4. Possible ways to extend a forest in $\mathcal{F}_{2n}$ to a spanning tree in $\mathcal{T}_{2(n+1)}$

Let $T_k = |\mathcal{T}_k|$ and $F_k = |\mathcal{F}_k|$. From the previous paragraphs, we obtain the recurrence

$$T_{2(n+1)} = 5T_{2n} + 3F_{2n}.$$

Since F_{2n} is featured in the above relation, it is equally important to keep track of F_{2n} . We can derive a recurrence for F_{2n} by similar reasoning. Starting from a spanning tree in $\mathcal{T}_{2n}$, there are three ways to append two vertices to get a forest in $\mathcal{F}_{2(n+1)}$:

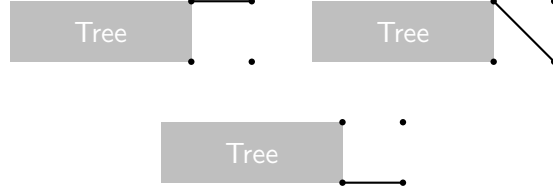


FIGURE 5. Possible ways to extend a spanning tree in $\mathcal{T}_{2n}$ to a forest in $\mathcal{F}_{2(n+1)}$

And there are two ways to extend a forest in $\mathcal{F}_{2n}$ to a forest in $\mathcal{F}_{2(n+1)}$:



FIGURE 6. Possible ways to extend a forest in $\mathcal{F}_{2n}$ to a forest in $\mathcal{F}_{2(n+1)}$

Thus, we get the recurrence

$$F_{2(n+1)} = 3T_{2n} + 2F_{2n}.$$

Our goal now is to combine the preceding recurrences to obtain a recurrence involving only T_{2n} . First, we let v_{2n} denote the column vector

$$v_{2n} = \begin{bmatrix} T_{2n} \\ F_{2n} \end{bmatrix}.$$

And if we define matrix A as

$$A = \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix},$$

we satisfy

$$v_{2(n+1)} = Av_{2n}.$$

Since the characteristic polynomial for A is

$$X_A(\lambda) = \lambda^2 - 7\lambda + 1,$$

by applying the Cayley-Hamilton Theorem, we obtain

$$A^2 - 7A + I = 0,$$

which can be rewritten as

$$A^2 = 7A - I.$$

Multiplying v_{2n} on both sides, we obtain

$$v_{2(n+2)} = 7v_{2(n+1)} - v_{2n}.$$

Hence:

$$(1) \quad \begin{aligned} T_{2(n+2)} &= 7T_{2(n+1)} - T_{2n}, \\ \text{and } F_{2(n+2)} &= 7F_{2(n+1)} - F_{2n}. \end{aligned}$$

From enumerating all possible spanning trees in $\mathcal{T}_2$ and $\mathcal{T}_4$ and all possible spanning forests in $\mathcal{F}_2$ and $\mathcal{F}_4$, we find the starting conditions:

$$(2) \quad \begin{aligned} T_2 &= 1, & T_4 &= 8, \\ \text{and } F_2 &= 1, & F_4 &= 5. \end{aligned}$$

□

2.2. Counting spanning trees of a triangular strip lattice with an odd number of vertices. We also consider triangular strip lattices with an odd number of vertices, since enumerating their spanning trees is a necessary prerequisite for counting balanced spanning trees in Section 3. Let Γ_{2n-1} be obtained from $\Gamma_{2(n-1)}$ by adding a vertex to the right end of the lattice and connecting it to the two rightmost vertices (see Figure 7).

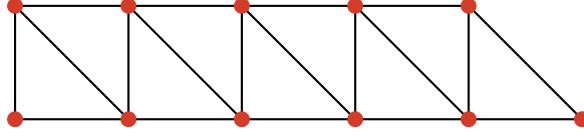


FIGURE 7. The graph of Γ_{11} , obtained by connecting a new vertex to the two rightmost vertices of Γ_{10} .

Let T_{2n-1} denote the number of spanning trees of Γ_{2n-1} . The following theorem gives a recurrence for T_{2n-1} .

Theorem 2.2 (Number of spanning trees in Γ_{2n-1}). *The number of spanning trees in Γ_{2n-1} , denoted by T_{2n-1} satisfies the recurrence relation:*

$$T_{2n+3} = 7T_{2n+1} - T_{2n-1}.$$

with initial conditions $T_3 = 3$ and $T_5 = 21$.

Proof. Since the orientation of the diagonal edge is determined by the structure of Γ_{2k-1} , we may assume without loss of generality that the diagonal joins the upper-left and lower-right vertices. The opposite orientation is symmetric and therefore yields the same number of spanning trees.

Now, we follow the counting technique as the one presented in the last section with some minor changes. We also use the recurrence relation established in the last section to make our work easier. As before, we denote the rightmost vertices of $\Gamma_{2(n-1)}$ by $\{u, v\}$. To get a spanning tree in Γ_{2n-1} , we can either expand from a spanning tree in $\Gamma_{2(n-1)}$ or expand from a forest in $\Gamma_{2(n-1)}$ consisting of two trees each containing a distinct vertex in $\{u, v\}$. Observe from Figure 8 that there are only two ways to get a spanning tree in Γ_{2n-1} from a spanning tree in $\Gamma_{2(n-1)}$ and only one way from a forest in $\Gamma_{2(n-1)}$.

So, from this, we obtain the following recurrence relation:

$$T_{2n+1} = 2T_{2n} + F_{2n}.$$

Using the recurrence relation from (1) and making the appropriate substitutions, we get the following:

$$(3) \quad T_{2n+3} = 7T_{2n+1} - T_{2n-1}$$

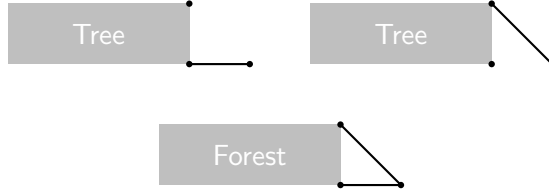


FIGURE 8. The possible ways to obtain a spanning tree in Γ_{2n-1} .

Also, from (2), we can find the initial values T_3 and T_5 .

$$(4) \quad T_3 = 2T_2 + F_2 = 2(1) + 1 = 3 \quad \text{and} \quad T_5 = 2T_4 + F_4 = 2(8) + 5 = 21,$$

which completes our proof. $\square$

Remark 2.3. Although Wang [Wan23] previously derived a recurrence for counting spanning trees in chains of triangles, a broader family of graphs that includes triangular strip lattices, we independently present a self-contained derivation specialized to triangular strip lattices. Although our techniques differ, the recurrences for triangular strip lattices in this section agree with the recurrence established by Wang [Wan23]:

$$g_t = 3g_{t-1} - g_{t-2},$$

where g_t denotes the number of spanning trees in a triangular strip lattice with t triangles. The proof of equivalence can be obtained by substituting $g_t = T_{t+2}$.

3. COUNTING BALANCED SPANNING TREES

3.1. Counting balanced spanning trees. With a clear recurrence relation established for the total number of spanning trees, we now calculate the number of balanced spanning trees in a triangular strip lattice. Our derivation follows the general framework introduced by Gallagher and Tapp [GT25], adapted to the geometry of triangular strip lattices.

To compute the total number of balanced spanning trees, our derivation proceeds in three main stages, as illustrated in Figure 9.

First, we find the number of certain “valid” loops of the *dual graph* Γ_{2n}^* (see Definition 2) by counting the *balanced partitions* of Γ_{2n} . Each such loop separates the vertices of Γ_{2n} , as shown in red and purple in Figure 9. These loops will give rise to candidate balanced edges (bordered in orange). Next, we observe that replacing either component (in red or purple vertices) with any spanning tree on the same vertex set and connecting them with any balanced edge yields a valid balanced tree. Finally, we use this to generate all possible balanced spanning trees.

For example, for the loop in Figure 9 we have a total of $3T_6^2$ balanced spanning trees associated with it as we have 3 possibilities for the balanced edge and T_6 possibilities for the spanning tree in either component.

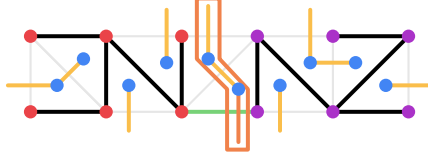


FIGURE 9. Overview of the three-step framework for counting the total number of balanced spanning trees.

We now provide a detailed derivation that encompasses all three stages. We begin by defining the dual graph of Γ_{2n} .

Definition 2 (Dual graph). The dual graph of Γ_{2n} , denoted by Γ_{2n}^* , is the graph whose vertices correspond to the faces of Γ_{2n} , including the unbounded exterior face, and whose edges correspond to the edges of Γ_{2n} . Specifically, each edge $e \in E(\Gamma_{2n})$ has a corresponding dual edge $e^* \in E(\Gamma_{2n}^*)$ connecting the two faces of Γ_{2n} adjacent to e (See Figure 10).

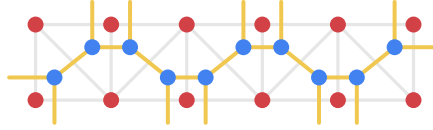


FIGURE 10. Dual graph of a triangular strip lattice. The vertices of Γ_{2n} are shown in red, the edges are grey, while the vertices of Γ_{2n}^* are shown in blue and the edges are in yellow. The vertex v_{∞}^* is omitted for clarity.

There is a one-to-one correspondence between the spanning trees of Γ_{2n} and the spanning trees of its dual Γ_{2n}^* . For any spanning tree $\mathcal{T} \in \mathcal{T}_{2n}$, its corresponding spanning tree $\mathcal{T}^* \in \mathcal{T}_{2n}^*$ consists precisely of edges $e^* \in \Gamma_{2n}^*$ that cross edges $e \in \Gamma_{2n} \setminus \mathcal{T}$. See Figure 11.

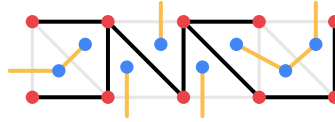


FIGURE 11. A spanning tree of Γ_{10} (in red and black) and the corresponding spanning tree in Γ_{10}^* (in yellow and blue) for the dual graph (v_{∞}^* not shown explicitly).

Under this formulation, removing a single edge $e \in E(\mathcal{T})$ to form a forest $\mathcal{T} \setminus \{e\}$ is equivalent to adding its corresponding edge e^* to $\mathcal{T}^*$, which creates a unique valid loop γ in $\mathcal{T}^* \cup \{e^*\}$. (See Figure 12.)

Let $\mathcal{T} \in \mathcal{T}_{2n}$ be a balanced spanning tree, and let $e \in E(\mathcal{T})$ be its balanced edge whose removal partitions $V(\Gamma_{2n})$ into two components of equal size. The removal of e induces the addition of $e^* \in \Gamma_{2n}^*$ and creates a unique cycle γ in $\mathcal{T}^* \cup \{e^*\}$. This cycle passes through the vertex v_{∞}^* , and we call it a *valid loop*.

Now, we claim the following:

Claim 3.1 (Number of valid loops). Γ_{2n} has n valid loops.

Proof. Observe that, when a valid loop is formed by the removal of the balanced edge, the vertices of Γ_{2n} are partitioned into two sets, each having n vertices (see Figure 12 for the case where $n = 5$). We define a *balanced partition* as a partition of the vertices of Γ_{2n} into two disjoint, sets having n vertices. For convenience, denote by L and P the set of valid loops and balanced partitions respectively. There is a bijective correspondence between valid loops and balanced partitions. Therefore, L and P have the same cardinality. We now count the number of balanced partitions.

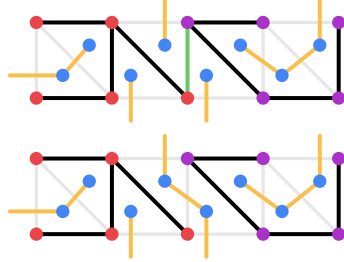


FIGURE 12. The removal of the balanced edge in Γ_{10} (green edge at top) creates a valid loop in the dual Γ_{10}^* that gives a balanced partition of the vertex set of Γ_{10} .

Let the two sets of vertices we get in a balanced partition be A and B , colored by red and purple respectively in Figure 12 and Figure 13. We can get a balanced partition by taking in A , the leftmost k vertices from the top row and leftmost $n - k$ vertices from the bottom row where $k \in \{1, 2, 3, \dots, n - 1, n\}$ and including the rest of the vertices of Γ_{2n} in B .

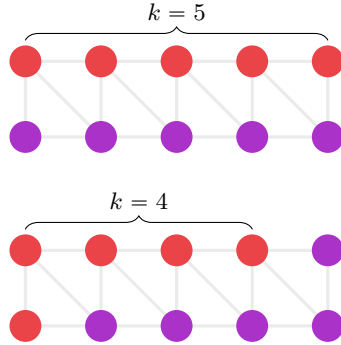


FIGURE 13. Two balanced partitions corresponding to $k = 5$ and $k = 4$ respectively.

From this, we can see that there are a total of n balanced partitions. Since L and P have the same cardinality, and P has n elements, L must also have n elements.

This proves our claim. $\square$

Next, following the indexing convention of Gallagher and Tapp [GT25], we label the n partitions as shown in Figure 14. Define $m = \lfloor n/2 \rfloor$. The indexing is chosen so that there are m indexed

partitions, and reflection across the vertical centerline pairs most partitions. Specifically, the non-symmetric partitions $1, \dots, m-1$ for even n and $0, \dots, m-1$ for odd n are paired with their reflections. For convenience, indexed partitions are precisely those for which the left side of the partition contains at least as many bottom-row vertices as top-row vertices.

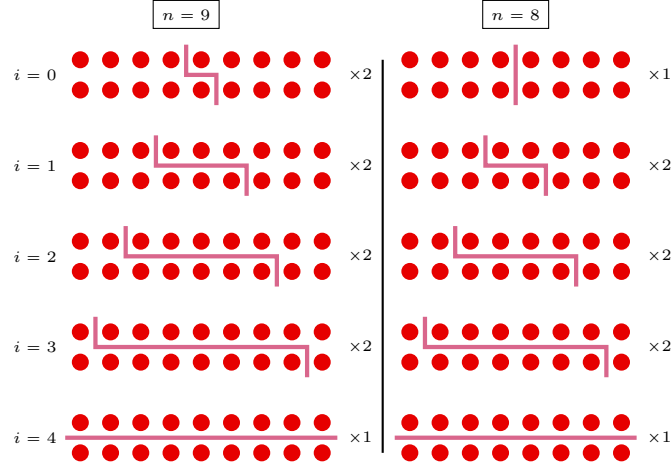


FIGURE 14. The n possible ways to partition the vertices of Γ_{2n} equally for $n = 9$ and $n = 8$. Inspired by [GT25].

By Claim 3.1, each partition corresponds to a valid loop in Γ_{2n}^* . We now use γ_i to denote the loop that corresponds to the indexed partition i , and μ_i to denote the loop that corresponds to the reflection of partition i . Note that since diagonals of faces break the symmetry that exists in a grid graph, γ_i does not necessarily have the same length as μ_i . Therefore, it is important to distinguish them.

Now, we will use Figure 15 to illustrate our strategy of counting the number of balanced spanning trees associated with each valid loop. Each valid loop divides the vertex set of the graph into two equal sets. A balanced spanning tree can be obtained by first forming a spanning tree on either side of the loop, then connecting the two trees by replacing one edge of the valid loop with the edge it intersects in the graph Γ_{2n} . The number of spanning trees on either side of the loop equals the number of spanning trees in *end blocks* (highlighted in yellow) of their respective sides. Therefore, if we denote the number of spanning trees in the left and right end blocks by T_L and T_R respectively, then each valid loop is associated with $\text{length}(\gamma_i)T_L T_R$ balanced spanning trees. Note that γ_i can be interchanged with μ_i here.

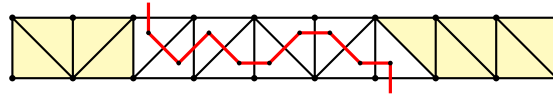


FIGURE 15. Example of a valid loop

Observe that both the length of the loop and the sizes of the end blocks depend only on the types of diagonals in the leftmost and rightmost squares that the loop passes through. For convenience, we call them *start* and *end squares*. This observation motivates the following structural parameter.

Definition 3 (Structural parameter σ_i). Given a 2-by- n triangular strip lattice, set $m = \lfloor n/2 \rfloor$. Let the type of diagonals that connect the top-left and bottom-right vertices of a square be denoted by “\,” and the type of diagonals that connect the top-right and bottom-left vertices of the square be denoted by “/.” Define σ_i as a parameter that encodes the symmetry of the diagonals in the start and end squares of the loop γ_i . If the $(m-i)$ th square and the $(n-m+i)$ th square have the same type of diagonals, i.e., (\, \) or (/, /), then $\sigma_i = 1$. Otherwise, if the squares have different types of diagonals, i.e., (\, /) or (/, \), then $\sigma_i = 0$.

Using the indexing systems and the defined structural parameter, we obtain the following formulae.

Theorem 3.2. Let S_{2n} denote the number of balanced spanning trees in Γ_{2n} . If n is even, i.e., $n = 2m$ for some m , then

$$S_{2n} = 2n - 1 + 3T_{2m}^2 + \sum_{i=1}^{m-1} [\sigma_i((4i+1)T_{2(m-i)+1}^2 + (4i+3)T_{2(m-i)}^2) + (1-\sigma_i)(8i+4)T_{2(m-i)+1}T_{2(m-i)}].$$

If n is odd, i.e., $n = 2m+1$ for some m , then

$$S_{2n} = 2n - 1 + \sum_{i=0}^{m-1} [\sigma_i((4i+3)T_{2(m-i)+1}^2 + (4i+5)T_{2(m-i)}^2) + (1-\sigma_i)(8i+8)T_{2(m-i)+1}T_{2(m-i)}].$$

Example 3.3. An example of how the formulae work in practice is as follows. Consider the graph Γ_{20} in Figure 15. Since $n = 10$ is even, we apply the formula for n even. First, we compute

$$2n - 1 = 2(10) - 1 = 19.$$

Next, observing that $m = \lfloor n/2 \rfloor = 5$, and using the recurrence from Theorem 2.1 or equation 7 in the next section to compute T_{2m} , we obtain

$$3T_{2m}^2 = 3T_{10}^2 = 3(2584)^2 = 20,031,168.$$

Finally, we calculate the value associated with each index i in the summand. For $i = 1$, observe that the diagonals of the start square and end square are the same type, so $\sigma_1 = 1$. Substituting $\sigma_i = 1$ for index $i = 1$ in the summand, we obtain

$$\begin{aligned} (4i+1)T_{2(m-i)+1}^2 + (4i+3)T_{2(m-i)}^2 &= 5T_9^2 + 7T_8^2 \\ &= 5(987)^2 + 7(377)^2 \\ &= 5,865,748. \end{aligned}$$

For $i = 2$, notice that the diagonals in the start and end squares are different, so $\sigma_2 = 0$. Substituting this into the $i = 2$ term in the summand, we obtain

$$\begin{aligned} (8i+4)T_{2(m-i)+1}T_{2(m-i)} &= 20T_7T_6 \\ &= 20(144)(55) \\ &= 158,400. \end{aligned}$$

Repeat this process for the remaining indices: $i = 3$ gives 4704, and $i = 4$ gives 172. The total number of balanced spanning trees of Γ_{20} equals the sum of all calculated results, which is 26,060,211.

Proof of Theorem 3.2. We now prove Theorem 3.2.

Even Case. We first consider the even case where $n = 2m$.

We begin with the symmetric partitions $i = 0$ and $i = m$. For $i = 0$, $\text{length}(\gamma_0)$ is always 3, and γ_0 divides the graph into two end blocks, each consisting of $2m$ vertices. As illustrated in Figure 15, γ_0 therefore contributes $3T_{2m}^2$ balanced spanning trees. For $i = m$, the loop γ_m has length $2n - 1$. In addition, on each half of the m -th partition, there is only one possible spanning tree. Thus, the number of balanced spanning trees associated with γ_m is $2n - 1$. See Figure 16.

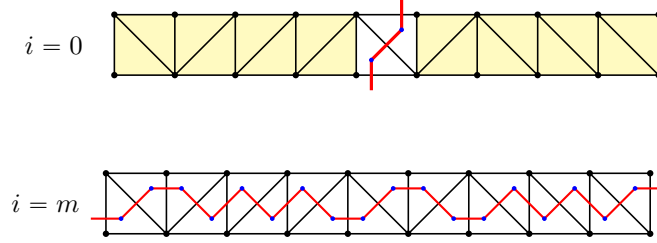


FIGURE 16. Example for $i = 0$ and $i = m$ in a 2-by-10 triangular strip lattice

For $i = 1, \dots, m - 1$, the contribution depends on the diagonal configuration. There are two cases: $\sigma_i = 1$ and $\sigma_i = 0$.

Recall that if $\sigma_i = 1$, the start and end squares of a valid loop contain the same type of diagonal. There are two possible configurations, namely $(\backslash, \backslash)$ and $(/, /)$. We consider the first configuration and determine the lengths of γ_i and μ_i together with the sizes of their end blocks.

Observe that a 2-by- n triangular strip lattice contains $n - 1$ squares. For $i \neq m$, both γ_i and μ_i pass through the start square $m - i$, the end square $n - m + i$, and every square between them. We refer to the squares strictly between the start and end squares as the *middle squares* (highlighted in pink in Figure 17). The number of the middle squares is $(n - m + i) - (m - i) - 1 = 2i - 1$. Every middle square contributes two edges to each valid loop, except for one of them, which contributes three edges. Hence the middle squares collectively contribute $2(2i - 1) + 1 = 4i - 1$ edges to each valid loop. Since γ_i does not intersect the diagonals of the start and end squares in this configuration, the start and end squares each contribute one additional edge. Meanwhile, since μ_i intersects the diagonals of the start and end squares, the start and end squares each contribute two additional edges. From this paragraph, we obtain

$$\text{length}(\gamma_i) = 4i + 1,$$

$$\text{length}(\mu_i) = 4i + 3.$$

Next, observe that each end block (highlighted in yellow in Figure 17) contains vertices belonging to the first $m - i - 1$ complete squares. Since γ_i does not intersect the diagonals of the start and end squares, each of its end blocks contains an extra vertex. Therefore, the end blocks of γ_i contain $2(m - i) + 1$ vertices each, and the end blocks of μ_i contains $2(m - i)$ vertices each. Together, γ_i and μ_i contribute

$$(4i + 1)T_{2(m-i)+1}^2 + (4i + 3)T_{2(m-i)}^2$$

balanced spanning trees.

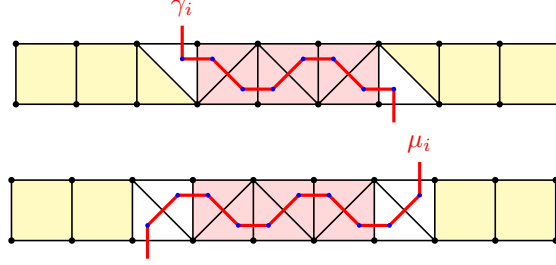


FIGURE 17. Example for $i = 2$ in a 2-by-10 triangular strip lattice ($\sigma_i = 1$)

In the other configuration $(/, /)$, the roles of γ_i and μ_i are simply reversed: γ_i has length $4i + 3$ and end blocks of size $2(m - i)$, while μ_i has length $4i + 3$ and end blocks of size $2(m - i)$. Hence, both configurations contribute the same number of balanced spanning trees.

For $\sigma_i = 0$, the possible configurations are $(/, \setminus)$ and $(\setminus, /)$. In either configuration, γ_i and μ_i each intersects only one of the diagonals of the start and end cells. (See Figure 18.) Therefore,

$$\text{length}(\gamma_i) = \text{length}(\mu_i) = 4i + 2.$$

Furthermore, both γ_i and μ_i have end blocks containing $2(m - i) + 1$ and $2(m - i)$ vertices. Therefore, $\sigma_i = 0$ contributes

$$(8i + 4)T_{2(m-i)+1}T_{2(m-i)}$$

balanced spanning trees.

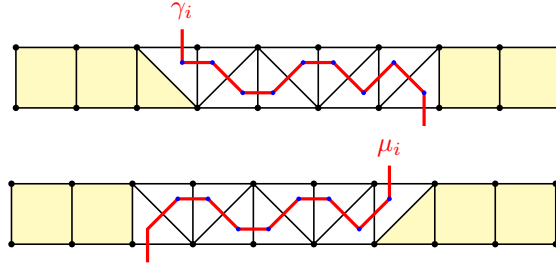
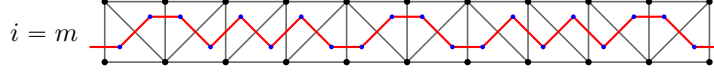


FIGURE 18. Example for $i = 2$ in a 2-by-10 triangular strip lattice ($\sigma_i = 0$)

Summing over all indices completes the proof for the even case. $\square$

Odd Case. The proof for the odd case is analogous to the even case. We first consider the symmetric partition $i = m$. Similar to the even case, $\text{length}(\gamma_m) = 2n - 1$, and on either side of γ_m , there is only one possible spanning tree. (See Figure 19.) Therefore, γ_m contributes $2n - 1$ balanced spanning trees.

FIGURE 19. Example for $i = m$ in a 2-by-11 triangular strip lattice

Next, we consider the cases when $\sigma_i = 1$ and $\sigma_i = 0$ for non-symmetric indices $0, 1, \dots, m-1$. Here, we apply the exact same strategy used in the even case. The only difference is that the middle squares here contribute $4i+1$ edges to each loop. Adjusting for this discrepancy and treating $i = 0$ as a non-symmetric partition, we achieve the formula for the odd case. $\square$

3.2. Maximizing and minimizing S_{2n} . We are also interested in figuring out the configurations for Γ_{2n} so that S_{2n} is maximized and minimized. Our strategy is to find a relationship between the parameter σ_i and the number of spanning trees associated with i . In doing so, we make the following observation.

Lemma 3.4. *For every non-symmetric partition pair i of Γ_{2n} , the number of balanced spanning trees contributed by i is maximized when $\sigma_i = 1$ and minimized when $\sigma_i = 0$.*

Proof. For even n and $1 \leq i \leq m-1$, the contributions corresponding to $\sigma_i = 1$ and $\sigma_i = 0$, are $(4i+1)T_{2(m-i)+1}^2 + (4i+3)T_{2(m-i)}^2$ and $(8i+4)T_{2(m-i)+1}T_{2(m-i)}$, respectively. Subtracting the latter from the former, we obtain

$$\begin{aligned} & (4i+1)T_{2(m-i)+1}^2 + (4i+3)T_{2(m-i)}^2 - (8i+4)T_{2(m-i)+1}T_{2(m-i)} \\ &= (4i+1)(T_{2(m-i)+1} - T_{2(m-i)})^2 - 2T_{2(m-i)}(T_{2(m-i)+1} - T_{2(m-i)}). \end{aligned}$$

Substituting the identity $T_{2(m-i)+1} = 2T_{2(m-i)} + F_{2(m-i)}$ yields

$$\begin{aligned} & (4i+1)(T_{2(m-i)} + F_{2(m-i)})^2 - 2T_{2(m-i)}(T_{2(m-i)} + F_{2(m-i)}) \\ &= 4(i-1)(T_{2(m-i)} + F_{2(m-i)})^2 + 3T_{2(m-i)}^2 + 8T_{2(m-i)}F_{2(m-i)} + 5F_{2(m-i)}^2. \end{aligned}$$

Since $i \geq 1$ and $T_{2(m-i)}, F_{2(m-i)} > 0$, the expression above is strictly positive. Hence,

$$(4i+1)T_{2(m-i)+1}^2 + (4i+3)T_{2(m-i)}^2 > (8i+4)T_{2(m-i)+1}T_{2(m-i)}.$$

This completes the proof for the even case.

The proof for odd n is analogous. For $0 \leq i \leq m-1$, subtracting the contribution corresponding to $\sigma_i = 0$ from the contribution corresponding to $\sigma_i = 1$, we obtain

$$(4i+3)T_{2(m-i)+1}^2 + (4i+5)T_{2(m-i)}^2 - (8i+8)T_{2(m-i)+1}T_{2(m-i)}.$$

Substituting $T_{2(m-i)+1} = 2T_{2(m-i)} + F_{2(m-i)}$ and following the same computation as in the even case yields

$$4i(T_{2(m-i)} + F_{2(m-i)})^2 + T_{2(m-i)}^2 + 4T_{2(m-i)}F_{2(m-i)} + 3F_{2(m-i)}^2.$$

This expression is also strictly positive. Hence, the contribution corresponding to $\sigma_i = 1$ is greater than that corresponding to $\sigma_i = 0$. $\square$

The corollary below immediately follows Lemma 3.4.

Corollary 3.5. S_{2n} is maximized when $\sigma_i = 1$ for all i and minimized when $\sigma_i = 0$ for all i .

4. CALCULATING THE PROPORTION OF BALANCED SPANNING TREES

To calculate the proportion of balanced spanning trees, we first figure out a closed-form formula for T_k , the number of spanning trees in a graph with k vertices. We have 2 cases depending on the parity of k .

If $k = 2n$, using (1) and (2), we get the following:

$$(5) \quad T_{2n} = \beta r^n - \theta r^{-n} \quad \text{where} \quad \beta = \frac{3\sqrt{5}-5}{10}, \quad \theta = \frac{3\sqrt{5}+5}{10} \quad \text{and} \quad r = \frac{7+3\sqrt{5}}{2}.$$

Similarly, $k = 2n + 1$, using (3) and (4), we get the following:

$$(6) \quad T_{2n+1} = \alpha(r^n - r^{-n}) \quad \text{where} \quad \alpha = \frac{1}{\sqrt{5}} \quad \text{and} \quad r = \frac{7+3\sqrt{5}}{2}.$$

We can unify both equations and obtain the following:

$$(7) \quad T_k = \alpha(\phi^{2k-2} - \phi^{-(2k-2)}) \quad \text{where} \quad \alpha = \frac{1}{\sqrt{5}}, \quad \phi = \frac{\sqrt{5}+1}{2}.$$

4.1. Finding the upper bound. The main goal of this section is to prove the following:

Proposition 4.1. *For any $n \geq 1$, if $\sigma_i = 1$ for all $i \in \{1, 2, \dots, m-1\}$, the following holds:*

$$\lim_{n \rightarrow \infty} \frac{S_{2n}}{T_{2n}} = 0.668328.$$

We deal with the case where $\sigma_i = 1$ for all $i \in \{1, 2, 3, \dots, m-1\}$. Recall by Corollary 3.5 that this gives the maximum S_{2n} . By Theorem 3.2, in our case, if $n = 2m$, then,

$$S_{2n} = (2n-1) + 3T_{2m}^2 + \sum_{i=1}^{m-1} ((4i+1)T_{2(m-i)+1}^2 + (4i+3)T_{2(m-i)}^2).$$

Observe that,

$$\begin{aligned} \sum_{i=1}^{m-1} ((4i+1)T_{2(m-i)+1}^2 + (4i+3)T_{2(m-i)}^2) &= \sum_{i=1}^{m-1} (4i+1)T_{2(m-i)+1}^2 + \sum_{i=1}^{m-1} (4i+3)T_{2(m-i)}^2 \\ &= \sum_{j=2; j \text{ is even}}^{2(m-1)} (2j+1)T_{n+1-j}^2 + \sum_{j=3; j \text{ is odd}}^{2m-1} (2j+1)T_{n+1-j}^2 \\ &= \sum_{i=2}^{n-1} (2i+1)T_{n+1-i}^2. \end{aligned}$$

And if $n = 2m + 1$ is odd, then,

$$S_{2n} = (2n-1) + \sum_{i=0}^{m-1} ((4i+3)T_{2(m-i)+1}^2 + (4i+5)T_{2(m-i)}^2).$$

Observe that,

$$\begin{aligned}
\sum_{i=0}^{m-1} ((4i+3)T_{2(m-i)+1}^2 + (4i+5)T_{2(m-i)}^2) &= \sum_{i=0}^{m-1} (4i+3)T_{2(m-i)+1}^2 + \sum_{i=0}^{m-1} (4i+5)T_{2(m-i)}^2 \\
&= \sum_{j=2; j \text{ is even}}^{2m} (2j+1)T_{n+1-j}^2 + \sum_{j=1; j \text{ is odd}}^{2m-1} (2j+1)T_{n+1-j}^2 \\
&= \sum_{i=1}^{n-1} (2i+1)T_{n+1-i}^2.
\end{aligned}$$

Now, with some substitutions, we can unify both formulae to obtain the following:

$$S_{2n} = (2n-1) + \sum_{i=1}^{n-1} (2i+1)T_{n+1-i}^2.$$

This equation applies to both even and odd n .

Proof of Proposition 4.1. Observe that, for $i \in \{1, 2, \dots, n\}$,

$$\begin{aligned}
(8) \quad \frac{T_{n+1-i}^2}{T_{2n}} &\stackrel{(7)}{=} \frac{\alpha^2(\phi^{4n-4i} + \phi^{-(4n-4i)} - 2)}{\alpha(\phi^{4n-2} - \phi^{-(4n-2)})} \\
&= \alpha \frac{\phi^{-4i} + \phi^{-8n+4i} - 2\phi^{-4n}}{\phi^{-2} - \phi^{-8n+2}}.
\end{aligned}$$

Also, note the following inequality:

$$(9) \quad T_{a+b} > 3T_a \cdot T_b.$$

This can be shown algebraically by using (7) or by noting that, if we consider two spanning trees $\mathcal{T}_a$ and $\mathcal{T}_b$ and we align them to form a forest, there are 3 ways to add an edge to get a spanning tree $\mathcal{T}_{a+b}$. Furthermore, there are more ways of getting a spanning tree $\mathcal{T}_{a+b}$ without joining two trees. This works whenever one of a or b is even when we place the spanning tree with even vertices on the left. See Figure 20 for an example.

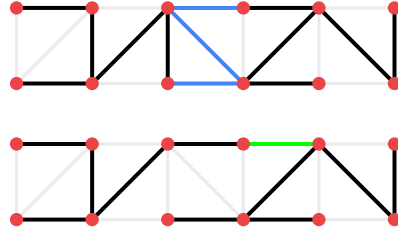


FIGURE 20. Top: Two spanning trees in Γ_6 (with black edges) that can be connected in 3 ways (the blue edges) to form a spanning tree in Γ_{12} . Bottom: A way to get a spanning tree in Γ_{12} without joining two trees (the edges in the left 6 vertices don't form a spanning tree for Γ_6).

When both a and b are odd, we can align them to create a graph in Γ_{a+b} . We still get three ways to create a spanning tree in Γ_{a+b} . Furthermore, there are more ways of getting a spanning tree $\mathcal{T}_{a+b}$ without joining two trees in $\mathcal{T}_a$ and $\mathcal{T}_b$. See Figure 21 for an example.

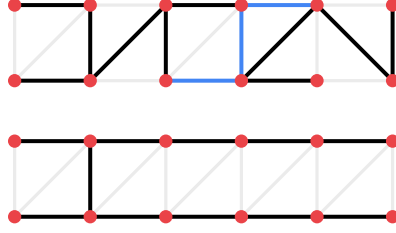


FIGURE 21. Top: Two spanning trees in Γ_7 on the left and Γ_5 on the right (with black edges) that can be connected in 3 ways (the blue edges) to form a spanning tree in Γ_{12} . Bottom: A way to get a spanning tree in Γ_{12} without joining two trees in Γ_7 and Γ_5 (the edges in the right 5 vertices don't form a spanning tree for Γ_5).

So in any case, this inequality holds. We can use this inequality to get an inequality that will be more useful to us:

$$(10) \quad T_{a+b+c} > 9T_a \cdot T_b \cdot T_c$$

Using this inequality, we get

$$(11) \quad T_{2n} = T_{(n+1-i)+(n+1-i)+(2i-2)} > 9T_{n+1-i}^2 T_{2i-2}.$$

Now, for $i \geq 4$, we get the following,

$$\begin{aligned} \frac{(2i+1)T_{n+1-i}^2}{T_{2n}} &\stackrel{(11)}{<} \frac{2i+1}{9T_{2i-2}} \\ &\stackrel{(7)}{=} \frac{2i+1}{9\alpha(\phi^{4i-6} - \phi^{-(4i-6)})} \\ &< \frac{\sqrt{5}}{9} \frac{2i+1}{(\frac{3}{2})^{4i-6}} = \frac{\sqrt{5}}{9} (2i+1) \left(\frac{2}{3}\right)^{4i-6}. \end{aligned}$$

With this fact, we can use the dominated convergence theorem to establish the following:

$$(12) \quad \lim_{n \rightarrow \infty} \sum_{i=1}^{n-1} \frac{(2i+1)T_{n+1-i}^2}{T_{2n}} = \sum_{i=1}^{\infty} \lim_{n \rightarrow \infty} \frac{(2i+1)T_{n+1-i}^2}{T_{2n}} = \sum_{i=1}^{\infty} (2i+1) \lim_{n \rightarrow \infty} \frac{T_{n+1-i}^2}{T_{2n}}.$$

where $T_k = 0$ if $k < 0$. Now, observe that,

$$(13) \quad \lim_{n \rightarrow \infty} \frac{T_{n+1-i}^2}{T_{2n}} \stackrel{(8)}{=} \lim_{n \rightarrow \infty} \alpha \frac{\phi^{-4i} + \phi^{-8n+4i} - 2\phi^{-4n}}{\phi^{-2} - \phi^{-8n+2}} = \alpha \frac{\phi^{-4i}}{\phi^{-2}} = \alpha \frac{\phi^2}{\phi^{4i}}.$$

Now, we get the following

$$\begin{aligned}
 (14) \quad \lim_{n \rightarrow \infty} \sum_{i=1}^{n-1} \frac{(2i+1)T_{n+1-i}^2}{T_{2n}} &\stackrel{(12),(13)}{=} \alpha \phi^2 \sum_{i=1}^{\infty} \frac{2i+1}{\phi^{4i}} \\
 &= \frac{\alpha \phi^2 (3\phi^{-4} - \phi^{-8})}{(1 - \phi^{-4})^2} \\
 &= \frac{10 + 3\sqrt{5}}{25} = 0.668328.
 \end{aligned}$$

Finally, observing that

$$\lim_{n \rightarrow \infty} \frac{2n-1}{T_{2n}} = 0,$$

we get

$$\lim_{n \rightarrow \infty} \frac{S_{2n}}{T_{2n}} = \lim_{n \rightarrow \infty} \frac{2n-1}{T_{2n}} + \lim_{n \rightarrow \infty} \sum_{i=1}^{n-1} \frac{(2i+1)T_{n+1-i}^2}{T_{2n}} = 0.668328.$$

This completes our proof. □

Remark 4.2. It is worth noting that if Γ_{2n} has $\sigma_i = 1$ for all i , this limit already agrees with $\frac{S_{2n}}{T_{2n}}$ in three decimal places for $n \geq 6$. So, for a fixed n , we can approximate the proportion of balanced spanning trees in triangular lattices where $\sigma_i = 1$ for all i very accurately with this limit.

4.2. Finding the lower bound. The main goal of this section is to prove the following two propositions:

Proposition 4.3. *For any even $n \geq 1$, if $\sigma_i = 1$ for all $i \in \{1, 2, \dots, m-1\}$, the following holds:*

$$\lim_{n \rightarrow \infty} \frac{S_{2n}}{T_{2n}} = 0.63087.$$

Proposition 4.4. *For any odd $n \geq 1$, if $\sigma_i = 1$ for all $i \in \{1, 2, \dots, m-1\}$, the following holds:*

$$\lim_{n \rightarrow \infty} \frac{S_{2n}}{T_{2n}} = 0.54493.$$

Proof for Proposition 4.3. By Theorem 3.2, if $n = 2m$, we get,

$$S_{2n} = 2n - 1 + 3T_{2m}^2 + \sum_{i=1}^{m-1} [(8i+4)T_{2(m-i)+1}T_{2(m-i)}].$$

Observe that, we have,

$$\begin{aligned}
\frac{(8i+4)T_{2(m-i)+1}T_{2(m-i)}}{T_{4m}} &\stackrel{(10)}{<} \frac{8i+4}{9T_{4i-1}} \\
&= \frac{8i+4}{9\alpha(\phi^{8i-3} - \phi^{-(8i-3)})} \\
&< \frac{8i+4}{9\alpha(\frac{3}{2})^{8i-3}} \\
&= \frac{\sqrt{5}}{9}(8i+4)\left(\frac{2}{3}\right)^{8i-3}.
\end{aligned}$$

With this fact, we can use the dominated convergence theorem to establish the following:

$$(15) \quad \lim_{m \rightarrow \infty} \sum_{i=1}^{m-1} \frac{(8i+4)T_{2(m-i)+1}T_{2(m-i)}}{T_{4m}} = \sum_{i=1}^{\infty} (8i+4) \lim_{m \rightarrow \infty} \frac{T_{2(m-i)+1}T_{2(m-i)}}{T_{4m}}.$$

Observe that

$$\lim_{m \rightarrow \infty} \frac{T_{2(m-i)+1}T_{2(m-i)}}{T_{4m}} = \lim_{m \rightarrow \infty} \alpha \frac{\phi^{-8i-2} - \phi^{-8m+2} - \phi^{-8m-2} + \phi^{-8(2m-i)+2}}{\phi^{-2} - \phi^{-(16m-2)}} = \frac{\alpha}{\phi^{8i}}.$$

Now, with this, we get

$$\begin{aligned}
(16) \quad \lim_{m \rightarrow \infty} \sum_{i=1}^{m-1} \frac{(8i+4)T_{2(m-i)+1}T_{2(m-i)}}{T_{4m}} &\stackrel{(15)}{=} \sum_{i=1}^{\infty} \alpha \frac{8i+4}{\phi^{8i}} \\
&= \alpha \left[\frac{8\phi^{-8}}{(1-\phi^{-8})^2} + \frac{4\phi^{-8}}{1-\phi^{-8}} \right] \\
&= 0.11841.
\end{aligned}$$

Now, also observe that,

$$(17) \quad \lim_{m \rightarrow \infty} \frac{3T_{2m}^2}{T_{4m}} \stackrel{(7)}{=} \lim_{m \rightarrow \infty} 3\alpha \frac{\phi^{-4} + \phi^{-(16m-4)} - 2\phi^{-8m}}{\phi^{-2} - \phi^{-(16m-2)}} = \frac{3\alpha}{\phi^2} = 0.51246.$$

So, we obtain,

$$\lim_{m \rightarrow \infty} \frac{S_{4m}}{T_{4m}} = \lim_{m \rightarrow \infty} \frac{4m-1}{T_{4m}} + \lim_{m \rightarrow \infty} \frac{3T_{2m}^2}{T_{4m}} + \lim_{m \rightarrow \infty} \sum_{i=1}^{m-1} \frac{(8i+4)T_{2(m-i)+1}T_{2(m-i)}}{T_{4m}} \stackrel{(16),(17)}{=} 0.63087.$$

□

Proof for Proposition 4.4. By Theorem 3.2, we establish that

$$\lim_{n \rightarrow \infty} \frac{S_{2n}}{T_{2n}} = \lim_{n \rightarrow \infty} \left(\frac{2n-1}{T_{2n}} + \sum_{i=0}^{m-1} \frac{(8i+8)T_{2(m-i)+1}T_{2(m-i)}}{T_{2n}} \right).$$

Again, since

$$\lim_{n \rightarrow \infty} \frac{2n-1}{T_{2n}} = 0,$$

we can eliminate the first term. From (10), we get

$$9T_{2(m-i)+1}T_{2(m-i)}T_{4i+1} < T_{4m+2} = T_{2n},$$

and consequently,

$$\frac{(8i+8)T_{2(m-i)+1}T_{2(m-i)}}{T_{2n}} < \frac{8i+8}{T_{4i+1}}.$$

Therefore, we can use the dominated convergence theorem to derive the following:

$$\lim_{n \rightarrow \infty} \frac{S_{2n}}{T_{2n}} = \sum_{i=0}^{\infty} (8i+8) \lim_{n \rightarrow \infty} \frac{T_{2(m-i)+1}T_{2(m-i)}}{T_{2n}}.$$

From 7, we can simplify the equation to

$$(18) \quad \lim_{n \rightarrow \infty} \frac{S_{2n}}{T_{2n}} = \sum_{i=0}^{\infty} (8i+8)\alpha\phi^{-8i-4} = \frac{8\alpha\phi^{-4}}{(1-\phi^{-8})^2} = 0.544933.$$

□

4.3. Random triangular strip lattices. Finally, we aim to determine the expected proportion of balanced spanning trees given a random configuration of the diagonals.

First, we precisely describe our random model. Let Γ_{2n} be a 2-by- n triangular strip lattice where the diagonal of each square face is chosen independently, such that it connects the top-left and bottom-right vertices with probability p , and the top-right and bottom-left vertices with probability $1-p$. As before, let S_{2n} denote the number of balanced spanning trees and T_{2n} denote the total number of spanning trees in Γ_{2n} .

We now state a more general version of Main Theorem 2:

Theorem 4.5 (Average Proportion for Triangular Strip Lattices). *Let Γ_{2n} be a random graph chosen as described above. As $n \rightarrow \infty$, the expected asymptotic proportion of balanced spanning trees is given by:*

Even Case ($n = 2m$):

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\frac{S_{2n}}{T_{2n}} \right] = \frac{9\sqrt{5}-15}{10} + (2p^2-2p+1) \frac{95-39\sqrt{5}}{50} + (2p-2p^2) \frac{210-82\sqrt{5}}{225}.$$

Odd Case ($n = 2m+1$):

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\frac{S_{2n}}{T_{2n}} \right] = (2p^2-2p+1) \frac{10+3\sqrt{5}}{25} + (2p-2p^2) \frac{68+28\sqrt{5}}{225}.$$

Proof. Recall that $\sigma_i = 1$ if the i th and $(n-i)$ th square have the same type of diagonal and $\sigma_i = 0$ otherwise. With standard calculations, we observe the following:

$$\mathbb{P}(\sigma_i) = \begin{cases} p^2 + (1-p)^2 & \text{when } \sigma_i = 1 \\ 2p(1-p) & \text{when } \sigma_i = 0. \end{cases}$$

From this, we see that,

$$(19) \quad \mathbb{E}[\sigma_i] = p^2 + (1-p)^2 = 2p^2 - 2p + 1 \quad \text{and} \quad 1 - \mathbb{E}[\sigma_i] = 2p - 2p^2.$$

With this, we proceed to the calculation of the expected value of $\frac{S_{2n}}{T_{2n}}$ as n limits to infinity. We deal separately with the odd and even cases.

Even Case. For $n = 2m$ using Theorem 3.2, (19), and linearity of expectation, we have the following:

$$(20) \quad \mathbb{E} \left[\frac{S_{2n}}{T_{2n}} \right] = \frac{4m-1}{T_{4m}} + \frac{3T_{2m}^2}{T_{4m}} + \sum_{i=1}^{m-1} [(2p^2 - 2p + 1)((4i+1)T_{2(m-i)+1}^2 + (4i+3)T_{2(m-i)}^2) + (2p - 2p^2)(8i+4)T_{2(m-i)+1}T_{2(m-i)}].$$

So, from (20), (14), (17), and (16) and standard limit calculations, we obtain the following:

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\frac{S_{2n}}{T_{2n}} \right] = \frac{9\sqrt{5} - 15}{10} + (2p^2 - 2p + 1) \frac{95 - 39\sqrt{5}}{50} + (2p - 2p^2) \frac{210 - 82\sqrt{5}}{225}.$$

□

Odd Case. For $n = 2m + 1$ using Theorem 3.2, (19), and linearity of expectation, we have the following:

$$(21) \quad \mathbb{E} \left[\frac{S_{2n}}{T_{2n}} \right] = \frac{4m+1}{T_{4m+2}} + \sum_{i=0}^{m-1} [(2p^2 - 2p + 1)((4i+3)T_{2(m-i)+1}^2 + (4i+5)T_{2(m-i)}^2) + (2p - 2p^2)(8i+8)T_{2(m-i)+1}T_{2(m-i)}].$$

So, from (21), (14), (18) and standard limit calculations, we obtain the following:

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\frac{S_{2n}}{T_{2n}} \right] = (2p^2 - 2p + 1) \frac{10 + 3\sqrt{5}}{25} + (2p - 2p^2) \frac{68 + 28\sqrt{5}}{225}.$$

□

A case of particular interest occurs when $p = \frac{1}{2}$, representing the scenario where the two possible diagonals in each square face of the $2 \times n$ grid are equally likely to be selected. In such a case, we obtain the following:

If $n = 2m$,

$$(22) \quad \lim_{n \rightarrow \infty} \mathbb{E} \left[\frac{S_{2n}}{T_{2n}} \right] = \frac{9\sqrt{5} - 15}{10} + \frac{95 - 39\sqrt{5}}{100} + \frac{210 - 82\sqrt{5}}{450} = 0.6496.$$

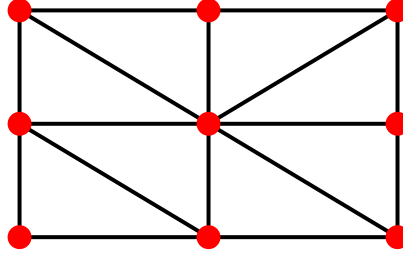
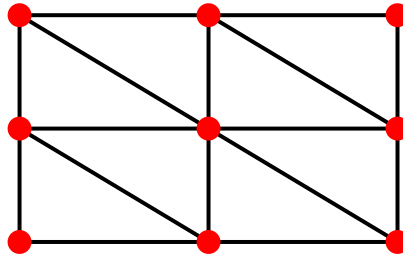
Similarly, if $n = 2m + 1$,

$$(23) \quad \lim_{n \rightarrow \infty} \mathbb{E} \left[\frac{S_{2n}}{T_{2n}} \right] = \frac{10 + 3\sqrt{5}}{50} + \frac{68 + 28\sqrt{5}}{450} = 0.6066.$$

It is worth noting that these are just the averages of their respective upper and lower bounds. □

5. CONCLUSION AND FUTURE DIRECTIONS

In this paper, we explored $2 \times n$ triangular strip lattices. Future research could focus on $k \times n$ triangular strip lattices where $k \geq 3$. Unlike the case for $k = 2$, when $k \geq 3$, the number of spanning trees in a triangular strip lattice depends on its diagonal configuration. For example, the triangular strip lattices in Figures 22 and 23 have the same k and n values, but their diagonal configurations are different. Therefore, they have different numbers of spanning trees (this can be verified with the help of Kirchhoff's Matrix Tree Theorem), complicating the computations in Section 4.

FIGURE 22. A 3×3 triangular strip lattice with 2146 possible spanning trees.FIGURE 23. A 3×3 triangular strip lattice with 2080 possible spanning trees.

A problem of interest would be to find the diagonal configuration that would result in the maximum and minimum number of spanning trees. In [Wan23], Wang found an upper and lower bound for the possible number of spanning trees T_n as n varies for the case where all the diagonals in the squares are of the same type (see Figure 23 for one of two possibilities for $n = 3$).

$$(32.4)^n \leq T_n \leq (81.3)^n.$$

We now discuss some complications that arise when calculating the number of balanced spanning trees in a particular $k \times n$ triangular strip lattice. For $k = 2$, we used dual loops to count the number of balanced spanning trees. Recall that the first step for doing so involves finding balanced partitions (see Claim 3.1) of vertices in $k \times n$ grid graphs. For $k = 2$, we had exactly n partitions which were straightforward to enumerate. For $k = 3$, the number of partitions is very complex. Specifically, the number of balanced partitions a_n , as derived in [Kte18], for a $3 \times 2n$ grid graph is:

$$\begin{aligned} a_n = & -3 + n - n^2 - \frac{1}{5}[(n-5)F_{3n+1} + (2n-1)F_{3n}] \\ & + \frac{1}{5} \sum_{m=0}^n \binom{2(n-m)}{n-m} [(3m+5)F_{3m+1} - (4m+3)F_{3m}] \end{aligned}$$

where F_k is, as before, the Fibonacci sequence. Due to this being a very complex sum, it is more difficult to enumerate the partitions which is a crucial step in counting the number of balanced spanning trees in both grid graphs and triangular strip lattices using our methods.

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